Department of Computer Science and Engineering

3^{rd} Year 2^{nd} Semester Final Examination' 2023 (Session: 2020-21)

Course Code: CSE 605 **Credit:** 3

Course Title: Software Engineering and Design Pattern

Time: 3 Hours **Total Marks:** 60

All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

Group A

[Answer any THREE sets of questions]

1. (a) Mention all the components of DFD with a proper diagram. 2

   (b) Explain the terms: Estimation management and Scheduling management. 3

   (c) Describe a real-world scenario where the Singleton design pattern is the best fit. Explain why this pattern is suitable in that context, and write the appropriate code to implement it. 5

2. (a) List the skills of a software project manager. 2

   (b) Suppose the main class of a software got too large, because the developer assigned too many responsibilities to that class. Which of the SOLID principles does that class violate? How should it be refactored? 3

   (c) Imagine you are the manager of a digital Biryani outlet, which allows customers to customize their own Biryani orders. Suppose your outlet offers two types of base Biryani – Chicken Biryani and Beef Biryani. Customers can choose any one base. You also allow your customers to enhance their Biryani with various add-ons such as Boiled Egg, Extra Meat, and Salad, where each add-on increases the total cost. At any time, a customer should be able to check what they are ordering (***description*** of the order) and the current total ***cost***. 5
   Which design pattern would you follow to maintain the orders in your Biryani outlet? Write an appropriate code.

3. (a) What is the main difference between Software Testing and Quality Assurance? 2

   (b) Briefly elaborate the building components of SCRUM and explain how they are interconnected. 3

   (c) A library keeps records of current loans of books to borrowers. Each borrower is' identified by borrower number and each copy of a book by an accession number. The information held about books is the title, author's name/s, publisher's name, publication date, international standard book number (ISBN - a unique book identifier), purchase price, classification (reference or fiction), and number of pages. A given book may be written by a number of different authors. A book may cover a number of different subjects. When a member of the library issues a borrowing request, s/he is granted it if the book is available and his/her personal borrowing restriction is not violated. Each member has a restriction on the maximum number of books to be allowed to borrow at a time depending on the type of membership, e.g., student/teacher. When a book is borrowed, the return date is automatically recorded based on the current date and the borrower's classification. Other borrowers, pending their return, may reserve books out on loan. Borrowers who hold overdue books or who have reached their loan limit, are flagged to prevent further borrowings. 5
   Design ER diagram to implement library system describe above.

4. (a) From the perspective of software engineering, explain the principle "Be Open to the Future" by answering the following questions. 2
   i. What measures should be taken to fulfill this principle?
   ii. What are the potential consequences if an organization fails to follow this principle?

   (b) Design a UML Class Diagram for an Online Banking System. Include classes such as Account, Customer, Transaction, and Loan with appropriate relationships (e.g., inheritance, association, aggregation). 4

   (c) To determine your eligibility for a job application in a tech company, you need to meet four conditions: **Relevant work experience > 2 years, Proficient in at least one programming language, complete a technical assessment**, and **submit a professional portfolio**. If any of these conditions are not fulfilled, your application **will be rejected**. Create an activity diagram illustrating the entire scenario, including **fork**, **join**, and **merge** conditions. 4

Group B

[Answer any THREE sets of questions]

5. (a) Your course teacher has assigned your group to prepare an SRS document for your SE project. Explain in detail what information should be included in the SRS to ensure it is complete and well-structured and identify any irrelevant information that should be excluded from the document. 3

   (b) Differentiate between Top down and bottom-up integration. 3

   (c) Write a note on 'Agile methodology' using a diagram. Mention the main advantage of the Agile model? 4

6. (a) Differentiate between White box testing and Black box testing. 2

   (b) Identify the type of risks indicated by the following examples: 3
   i. A software feature stops working properly when tested with real-world data.
   ii. The software cannot be sold in some countries due to new government rules.
   iii. A key project sponsor pulls out during development, affecting funding.

   (c) Draw a Use Case Diagram for an Online E-Learning Platform. Identify actors such as Student, Instructor, Administrator, and their corresponding use cases like Enroll in Course, Submit Assignment, Grade Assignment, and Manage Courses. 5

7. (a) Define Efficiency and Interoperability in terms of requirement engineering. 2

   (b) You are testing a university management system you developed, which includes features like student registration, grading, and fee payment. What key testing activities would you perform throughout the testing process to ensure the system works correctly, performs well, and is secure? Explain. 4

   (c) Your team is developing safety-critical software for a medical device. The development process strictly follows a structured approach where each design phase is paired with a corresponding testing phase. Draw this development process, clearly showing the connection between design and testing stages. Highlight its key advantages and disadvantages in the context of such a project. 4

8. (a) Briefly explain the three key activities under risk assessment: risk identification, risk analysis, and risk prioritization. 4

   (b) Write appropriate code for the following scenario: 6
   You have to design a navigation system for a ride-hailing application that supports multiple travel modes: **Driving** and **Walking**, each with unique methods for calculating the **travel time**, and **cost**. Create a ***NavigationContext*** class that allows the user to set or change the travel mode dynamically during runtime, using a ***TravelMode*** interface with methods like *calculateTravelTime()*, and ***estimateCost()***. Implement at least two concrete travel modes: **Driving** (consider fuel cost), **Walking** (assume zero cost). In the main class, show how the user can choose a travel mode and see the corresponding travel time and cost.